VNR Vignana Jyothi Institute of Engineering

Electronics and Communication Engineering | Course Code: EC202

Signals and Systems - Laboratory Manual & Practical Guide

Syllabus Focus: EC202 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Signal Classifications & System Properties

" Syllabus Scope: Continuous vs discrete time, energy vs power signals, linearity, time-invariance, causa

Unit 2: LTI Systems & Convolution

" Syllabus Scope: Convolution integral, convolution sum, impulse response, step response, cascade and par

EC202: Signals and Systems - Advanced Topics & Assessment

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Fourier Analysis of Continuous-Time Signals

" Continuous-Time Fourier Series CTFS, Dirichlet conditions, Continuous-Time Fourier Transform CTFT, duality,

Unit 4: Laplace Transform & System Stability

" Region of Convergence ROC properties, unilateral and bilateral Laplace transforms, transfer functions, pole-

Unit 5: Z-Transform & Discrete-Time Analysis

" Z-Transform definition, ROC properties, inverse Z-Transform partial fractions, residue, stability and causal

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Generation and Elementary Transformations of Continuous and Discrete Test Signals in Python

Lab 2: Numerical Verification of Convolution Integral and Convolution Sum

Lab 3: Fourier Spectrum Analysis of Periodic Square and Triangular Waveforms

Lab 4: Pole-Zero Mapping and Impulse Response Verification of LTI Systems in Z-Domain

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Alan V. Oppenheim, Alan S. Willsky - Signals and Systems Pearson 2nd Ed
* Simon Haykin, Barry Van Veen - Signals and Systems Wiley 2nd Ed

Authorized NoteVault Educational Document " Verified for Registered University Students